

Wearable AI For On-Device Frailty Assessment

Corresponding Author: Professor Philipp Gutruf

This file contains all reviewer reports in order by version, followed by all author rebuttals in order by version.

Version 0:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

This article is tackling a very important problem in the AI and wearables field which limits their integration into clinical practice.

- What are the noteworthy results?

The concept of a new wearable device for frailty measurement is proposed that has some advantages over existing wearables, mainly in terms of size and recharging mechanism.

- Will the work be of significance to the field and related fields? How does it compare to the established literature? If the work is not original, please provide relevant references.

The work is interesting, and the idea has some novelty over the state of art in the combination of a few new elements. For example, Canary Medical/Zimmer Biomet has a commercially available implantable knee sensor for joint arthroplasty which also charges by placing a device under the patient's bed without human intervention. OPUM Technologies has a wearable sensor with edge AI that does real-time gait analytics and exercise recognition to reduce the data transmitted back to the patient's phone via BLE in order to preserve battery life. It seems frailty measurement is also known in-clinic. So, the novelty would be in combining all these things, however it seems the paper does not actually report long-term frailty measurement, the only long-term operation data is with one healthy subject for 10 days.

- Does the work support the conclusions and claims, or is additional evidence needed?

To prove the device is actually able to do long term frailty monitoring, further results of this would need to be collected and presented. For example, the acceptability of a frail patient wearing the sensor is not tested, let alone over a period of time at home.

- Are there any flaws in the data analysis, interpretation and conclusions? Do these prohibit publication or require revision? Not that I could determine, although some more details on the specific gait and activity detection algorithms would be helpful

- Is the methodology sound? Does the work meet the expected standards in your field?

The methodology is sound, but does fall short in validating the proposed novelty of the research

- Is there enough detail provided in the methods for the work to be reproduced?

No, more details would be needed.

- Other comments

Include number of participants and other quantitative data in the abstract. What were the results? Did it work?

Intro: Authors say "...requiring no interaction by the wearer at any point, including during recharging and data collection"

However recharging and data collection are examples of user interaction. Where does the BLE go? How does that get back to the clinic? If the device is shipped how does the user setup the charger under their seat?

Pg 3. These statements have no references.. "On-device continuous step isolation and inference significantly reduces the amount of data transmitted by the device, conserving battery life, enhancing data privacy, and providing quantifiable long-term metrics with an overall accuracy greater than 90% when classifying between healthy and pre-frail steps. Studies with healthy and pre-frail subjects aged 65 and older demonstrate fidelity on-par with current clinical gold standard gait analysis devices, and showcase device ability to perform real time, exclusively on-device inference for continuous frailty status determination"

The authors say "The current clinical standard for frailty diagnosis, illustrated Fig. 1A, is reactionary rather than preventive, with diagnoses often only occurring after incidental fall or hospitalization." The new proposed method as per Figure 1 is also reactionary, as the gait assessment appears to happen after a fall and initial consultation. All this new method would do is remove one follow up with an in-clinic assessment, with an in-home assessment. If anything this actually delays diagnosis as at home measurement takes longer than in clinic assessment, and will no doubt have some usability and setup issues with elderly folks.

Authors should add some downsides of wearables to provide a balance, e.g. skin attachment, slipping, rubbing/comfort, social stigma etc

Results section

Gold standard. When were results taken from Bio sensors? If the goal is to show they can last for long periods then the data should be collected after say 60 days and/or with low charge, not when fully charged.

Methods

Seems like methods come after results section, doesn't make sense? Also some methods are in results section?

What is the functionality of the mesh fabrication, not clear what that is for and who it fits into the sensor design, manufacture and operation.

No clear description of electronics circuit and sensor design, i.e. controller specification/model, power storage, sensor architecture etc

No clear description of algorithms for step count, walking speed and sit to stand detection. I found some description but it's in the results section

Seems like the sensor parameters (BME rate) were adjusted from the gold standard testing to 10-day tests, this doesn't make the gold standard representative of at home use.

Reviewer #2

(Remarks to the Author)

1. The manuscript highlights its main contribution as enabling a transition from reactive, post-hospitalization frailty diagnosis to preventive, home-based diagnostics. The topic should be of interest to the readers of NC, and the paper is generally well written.

Some comments are provided below for consideration.

1. However, from Figure 1A it appears that patients are still prescribed the device only after a fall or hospitalization, following a clinical diagnosis. This raises a quick question: if such reactive clinical events still trigger device usage, how can the approach be truly considered preventive rather than reactive? The authors should consider clarifying under what circumstances the device would be deployed in practice, and whether there is a clear pathway for preventive adoption in at-risk but not yet hospitalized individuals.

2. In the model training process, the dataset contains a strong imbalance between healthy and pre-frail subjects (16 vs. 4). Such imbalance may risk biasing the classifier towards the majority class. It is recommended that the authors explore strategies to mitigate this issue, such as stratified sampling, data resampling, or synthetic minority oversampling methods (e.g., SMOTE), to avoid skewing results and to better validate the robustness of the model.

3. The work relies on a single 70/30 train-test split for model evaluation. While this provides an initial performance estimate, it does not fully rule out overfitting or ensure generalizability, especially given the small sample size. Cross-validation (e.g., leave-one-subject-out) or external validation would strengthen the claims regarding model robustness and clinical applicability.

4. Beyond cross-validation, the inclusion of learning curves would provide additional evidence regarding the model's bias-variance behavior and its capacity to generalize with more data.

5. Although the dataset includes a relatively large number of gait segments, these segments are not truly independent samples, as many originate from the same individuals. This could raise the risk that the model may be overfit to subject-specific characteristics, thereby limiting its generalizability to new individuals. Please discuss.

6. While the manuscript provides some waveform-level interpretations, the discussion remains largely algorithm-centric. A clearer articulation of how per-step inferences can be translated into clinically meaningful diagnostic indicators or longitudinal trend reports would strengthen the clinical relevance of the work. Please think about and discuss this.

7. The evaluation of the reported ~21% energy savings for on-device inference is confined to short-term, laboratory-based measurements. In real-world conditions across varying walking speeds, activity types, and usage durations, the device's overall power consumption may differ substantially. A more comprehensive energy analysis under daily living conditions is needed.

Version 1:

Reviewer comments:

Reviewer #1

(Remarks to the Author)

Thank you for the significant updates based on my comments.

While accepting that further work sits with commercialization efforts, I still believe that the study design and results do not strongly support the claimed novelty.

Reviewer #2

(Remarks to the Author)

The authors have made satisfactory revisions.

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Response to comments of Reviewer 1:

General comments:

This article is tackling a very important problem in the AI and wearables field which limits their integration into clinical practice.

What are the noteworthy results?

The concept of a new wearable device for frailty measurement is proposed that has some advantages over existing wearables, mainly in terms of size and recharging mechanism.

Our Response

We thank the reviewer for their comments and positive assessment of the manuscript's significance. We appreciate the detailed comments and have provided responses below.

Comment 1

Will the work be of significance to the field and related fields? How does it compare to the established literature? If the work is not original, please provide relevant references.

The work is interesting, and the idea has some novelty over the state of art in the combination of a few new elements. For example, Canary Medical/Zimmer Biomet has a commercially available implantable knee sensor for joint arthroplasty which also charges by placing a device under the patient's bed without human intervention. OPUM Technologies has a wearable sensor with edge AI that does real-time gait analytics and exercise recognition to reduce the data transmitted back to the patient's phone via BLE in order to preserve battery life. It seems frailty measurement is also known in-clinic. So, the novelty would be in combining all these things, however it seems the paper does not actually report long-term frailty measurement, the only long-term operation data is with one healthy subject for 10 days.

Our Response

We thank the reviewer for their comment, thorough comparison with clinically available technologies, and suggestion to highlight the novelty of this work. As mentioned, there are no technologies which can assess frailty outside of the clinic, which significantly limits diagnostics, especially in resource-constrained communities. This platform would be the first device to provide seamless diagnostic capacity for at-home frailty assessment. The OPUM technologies wearable sensor requires significant patient interaction to collect data using a phone application and to remove and replace the sensor for battery charging.

The novelty of the demonstrated device is in the minimal interaction needed by the wearer. 1. This technology utilizes far field wireless power transfer that allows the user to keep the device on 24/7 without removal for battery charging, critical for frail patient populations with limited mobility. 2. The device has a fundamentally different form factor than common brick-and-strap devices (like the OPUM device noted). Together with the wireless charging, this breathable, water- and sweatproof form factor enables the wearer to keep the device on for weeks without interacting with it. 3. The onboard AI allows for inference on device. This has multiple benefits: (i) It saves energy on the wearable, as shown by the measurements in Fig. 3C. (ii) It omits the need to transmit raw biosignals from the device to the cloud which can quickly result in large data amounts that are expensive to handle due to high bandwidth connectivity requirements, which can be challenging with a mobile phone as an access point, especially in rural communities. (iii) Inference data can be transferred directly into the Electronic Healthcare Record (EHR), making integration into standard of care and alerting systems much easier.

Conceptually we showcase the BEAD as a device for frailty assessment in this work, however, the device architecture, model training approach and on-device model deployment have broad

utility for long-term assessment of high data rate biosignals where machine learning can condense large data streams into clinically relevant, actionable metrics.

We have modified the manuscript to improve clarity and emphasize the novelty of the work.

Modifications to the Manuscript

Page 18 Line 346

Since the ~~biosymbiotic wearable device~~BEAD requires no interaction from the patient ~~beyond initial placement of the powercaster at a desk or bedside~~, is recharged wirelessly, and can be worn while bathing, the patient would simply wear the device on their thigh and continue with their normal routine.

Page 18 Line 349

During habitual gait, ~~i.e., step activity recognized by the BEAD's onboard IMU~~, the device wakes up, records ~~continuous angular velocity of the leg~~, isolates individual steps, and performs inference, aggregating the results over time and transmitting them off-device when possible, potentially integrating them directly into ~~EHRs~~ / electronic patient health information (ePHI)²⁸ or transmitting them directly to the telehealth provider.

Page 22 Line 423

Biosymbiotic wearable technologies with on-device ML offer substantial advantages over existing wearable devices, allowing patients in low-resource settings to receive clinical-grade frailty assessment. ~~The soft, breathable, lightweight form factor that defines biosymbiotic architecture is adhesive-free and tailored to conform to each patient for maximum comfort, allowing for weeks-long wear without irritation. Far-field power harvesting enables continuous operation without removal for charging (which is typically required by conventional brick-and-strap wearable devices), critical for potentially frail individuals with limited mobility. Exclusively on-device inference removes the requirement for constant connectivity that is often limited or unavailable, especially in rural communities, and eliminates the need for storing the large amounts of data on devices or costly cloud infrastructure, and allows for straightforward integration into EHRs.~~

Comment 2

Does the work support the conclusions and claims, or is additional evidence needed?

To prove the device is actually able to do long term frailty monitoring, further results of this would need to be collected and presented. For example, the acceptability of a frail patient wearing the sensor is not tested, let alone over a period of time at home.

Our Response

We thank the reviewer for their comment and concerns for device acceptability and practical use. We acknowledge that demonstrating long-term in-home wear by a frail patient would be a strong validation of the technology. However, user acceptability studies, clinical validation with statistical significance, and healthcare system integration (which would be required to carry out such studies efficiently) are typically part of commercialization efforts beyond exploratory technology demonstration such as the device presented in this work.

For context, considerations that frame the magnitude of such a study: given the population and demographics of Tucson, Arizona (547,000 general population with 17% age 65+)³⁹, the general incidence of frailty (15%)^{40,41}, and the number of frail subjects who are admitted to a hospital at least once in a year (42% frail with ≥ 1 hospital visit/year)⁴⁰, the subject pool is rapidly limited: $547,000 * 0.17 * 0.15 * 0.42 = 5,858$ patients

Factoring in the number of subjects who would benefit from intervention (40%)^{R1}, the ability to contact and recruit these subjects for a study (estimate ~30% of Tucson seen by all Banner University Medical Centers combined^{R2}, 44% willingness to enroll^{R3}), and general study retention (70%)^{R4}, $5,858 * 0.4 * 0.3 * 0.44 * .7 = 216$ potential test subjects, assuming exceptional enrollment efficiency and broad clinician subscription to the trial across the entire health system in Tucson.

It becomes clear that this clinical trial would need to be a multi-institution, multi-city effort. A trial that would yield many interesting results, improve the model, and could set the foundation of establishing and translating a new clinical tool, however, is well beyond this initial tech demonstration and we acknowledge that subject recruitment is a limitation of this study.

39. U.S. Census Bureau (2023). American Community Survey 1-year estimates. Retrieved from Census Reporter Profile page for Tucson, AZ. Accessed 09/18/2025.
<http://censusreporter.org/profiles/16000US0477000-tucson-az/>
40. Bandeen-Roche K, Seplaki CL, Huang J, Buta B, Kalyani RR, Varadhan R, Xue QL, Walston JD, Kasper JD. *Frailty in Older Adults: A Nationally Representative Profile in the United States*. J Gerontol A Biol Sci Med Sci. 2015 Nov;70(11):1427-34. doi: 10.1093/gerona/glv133. Epub 2015 Aug 21. PMID: 26297656; PMCID: PMC4723664.
41. Kim, D. H. & Rockwood, K. Frailty in Older Adults. N. Engl. J. Med. 391, 538–548 (2024).
- R1. Tze Pin Ng, Liang Feng, Ma Shwe Zin Nyunt, Lei Feng, Mathew Niti, Boon Yeow Tan, Gribson Chan, Sue Anne Khoo, Sue Mei Chan, Philip Yap, Keng Bee Yap, *Nutritional, Physical, Cognitive, and Combination Interventions and Frailty Reversal Among Older Adults: A Randomized Controlled Trial*, The American Journal of Medicine, Volume 128, Issue 11, 2015, Pages 1225-1236.e1, ISSN 0002-9343,
<https://doi.org/10.1016/j.amjmed.2015.06.017>.
- R2. American Medical Association. Residency & Fellowship Programs Database, Banner-University Medical Center Tucson, Banner-University Medical Center South.
<https://freida.ama-assn.org/institution/030506>. Accessed 09/08/2025.
- R3. Sood A, Prasad K, Chhatwani L, Shinozaki E, Cha SS, Loehrer LL, Wahner-Roedler DL. Patients' attitudes and preferences about participation and recruitment strategies in clinical trials. Mayo Clin Proc. 2009 Mar;84(3):243-7. doi: 10.4065/84.3.243. PMID: 19252111; PMCID: PMC2664601.
- R4. Teague, S., Youssef, G.J., Macdonald, J.A. et al. Retention strategies in longitudinal cohort studies: a systematic review and meta-analysis. BMC Med Res Methodol 18, 151 (2018).
<https://doi.org/10.1186/s12874-018-0586-7>

Page 23 Line 443

Realizing this clinical potential will require user acceptability studies and a multi-institution multi-site clinical trial with a significantly larger, diverse cohort; longitudinal in-home tracking of frailty progression is needed to create a generalizable, clinically relevant model. Nevertheless, the approach demonstrated here has the potential to overcome current operational, clinical, user and payer challenges by enabling seamless diagnostics with minimal interruption of daily routine and clinical practice.

Comment 3

Are there any flaws in the data analysis, interpretation and conclusions? Do these prohibit publication or require revision?

Not that I could determine, although some more details on the specific gait and activity detection algorithms would be helpful

Our Response

We thank the reviewer for their comment and suggestions to improve clarity surrounding gait and activity detection.

Activity recognition (still/sitting, walk, run) is performed by the BEAD's onboard inertial measurement unit (IMU). This IMU (Bosch, BM270) consists of a gyroscope, an accelerometer, and an internal controller that allows the user to enable or disable the gyroscope and/or accelerometer at any time. The internal controller embeds a proprietary decision tree model that continuously analyzes the accelerometer (which is always active for this use case) signal to classify/recognize activities. Activity recognition happens internal to the IMU and utilizes very little power (~72 μ W max), allowing the IMU to signal to the BEAD's microcontroller when the wearer has started or stopped walking.

Once step activity is detected, the microcontroller enables the IMU's gyroscope (which consumes nontrivial amounts of power, ~600 μ W average, and is normally disabled to conserve power when the wearer is still) and buffers incoming angular velocity data for ten seconds while the wearer walks around. Once the buffer is full, the microcontroller then performs peak finding algorithms, step extraction, data normalization, feature extraction, and inference, storing the classification results in non-volatile memory. This process continues until the IMU signals that the wearer has stopped moving, where the microcontroller then finishes analyzing any unprocessed steps, disables the gyroscope, and enters a low power mode.

We have amended the manuscript to better elaborate on these algorithms.

Modifications to the Manuscript

Page 10 Line 181

Using the BEAD's onboard IMU's (Bosch, BMI270) embeds a proprietary decision tree model that continuously evaluates accelerometer signals to detect motion and perform walking/running activity ~~detection~~ recognition, generating hardware-level interrupts on activity change. Using this interrupt allows the wearable device to enter a low-power mode that consumes 10x less power while maintaining quick wake timing (< 35 ms from low-power standby to normal operation, fig. S45) to collect real time gait data immediately once the wearer begins walking, with no additional effort or interaction required to begin data logging.

Page 18 Line 349

During habitual gait, i.e., step activity recognized by the BEAD's onboard IMU, the device wakes up, records continuous angular velocity of the leg, isolates individual steps, and performs inference, aggregating the results over time and transmitting them off-device when possible, potentially integrating them directly into EHRs / electronic patient health information (ePHI)²⁸ or transmitting them directly to the telehealth provider.

Modifications to the Supplementary Information (Methods)

Page 29 Line 574

On-device inference

When the BEAD's onboard IMU recognizes walking/running activity (through its proprietary embedded decision tree model analyzing its accelerometer signals), the IMU sends a hardware-level interrupt to the onboard microcontroller to wake it up. The microcontroller then enables the IMU's gyroscope and buffers ten seconds of 16-bit angular velocity data sampled at 200 Hz.

Individual step isolation is then performed by finding peaks in the angular velocity buffer (local maxima with at least 50 degrees per second angular velocity and ~1.1 seconds inter-peak spacing, parameters determined from the training dataset) and copying 218-sample step snippets centered at the index of each peak to a new buffer.

The sample times and angular velocity from a step are normalized [0, 1] and transformed using the minimized, embedded MINIROCKET to extract features. The features are then passed to the embedded random forest classifier, with the output class timestamped and stored in on-device non-volatile memory. All steps detected in the buffer are analyzed sequentially. During this time, the IMU's gyroscope remains active and continues to collect samples. Once the microcontroller has processed all steps, it retrieves any samples from the gyroscope and continues feature extraction and inference until the IMU detects that the wearer has stopped moving. The microcontroller then performs any last inferences, disables the IMU's gyroscope to save power, and enters a low power sleep until motion is detected again.

Comment 4

Is the methodology sound? Does the work meet the expected standards in your field?

The methodology is sound, but does fall short in validating the proposed novelty of the research

Our Response

We thank the reviewer for their comment and recognition of the methodology.

We want to emphasize that the primary novelty of this work lies primarily in the realization of on-device ML on a continuously wearable device and its demonstration as continuous diagnostic tool that does not require interaction by the wearer. This demonstration is a result of holistic system-level engineering with low power design and soft electronics that enable this demonstration. To highlight this novelty, we have added a table to the SI that compares existing technologies against the demonstrated device, Supplementary Table S2. Comparison with existing gait-based frailty assessment wearables.

Modifications to the Manuscript

Page 13 Line 233

The significance of this device compared to existing wearable technologies for frailty detection^{3,5,7,12,27,30–35}, see **Supplementary Table S2**, is clearly demonstrated by its ability to autonomously and efficiently evaluate a patient's gait in real time without any external infrastructure.

Modifications to the Supplementary Information (Figures) Supplementary Table S2

Category	Ref.	Year	Sample Device Image	Wearable type	Sensing target	ML type	On-device AI	Continuous operation
Real-time AI Analysis	[This Work]	-		Biosymbiotic mesh	Thigh angular velocity	RF	Y	Y
	[12]	2025		Brick-and-strap	Ankle angular velocity	RF	N	N
Batch processing AI analysis	[31]	2022		Commercial adhesive-based	6-axis wrists, ankles, waist IMU	SVM, DNN	N	N
	[32]	2022		Commercial brick-and-strap	3-axis wrist and low back acceleration	GNB	N	N
Statistical analysis only	[30]	2022		Commercial brick-and-strap	3-axis wrist acceleration	None	N/A	N
	[33]	2021		Brick-and-strap	6-axis ankle IMU	None	N/A	N
	[34]	2018		Commercial pendant necklace	3-axis chest acceleration	None	N/A	N
	[35]	2015		Commercial brick-and-strap	6-axis shanks, thighs, & torso IMU	None	N/A	N

Supplementary Table S2. Comparison with existing gait-based frailty assessment wearables. Comparison of BEADs with published literature utilizing wearable devices for gait-based frailty assessment. IMU = Inertial Measurement Unit (Accelerometer and Gyroscope). RF = Random Forest, SVM = Support Vector Machine, DNN = Deep Neural Networks, GNB = Gaussian Naive

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Comment 5

Is there enough detail provided in the methods for the work to be reproduced?

No, more details would be needed.

Our Response

We thank the reviewer for their comment and suggestion to provide more details to enhance reproducibility. We have revised the manuscript to better detail step / feature detection algorithms (see response to Comment 3 above) and model training (see response to Reviewer 2 Comments 3-5). We believe these revisions have significantly improved the reproducibility and clarity of the work and have copied them below for reference.

Modifications to the Manuscript

Page 10 Line 173

Onboard wireless far-field power harvesting **highlighted** using commercially available, FCC-approved power casting systems enable wireless recharging at-distance, such as at a work desk or during sleep, without wearer interaction.^{15,16} **These power casting systems plug into any wall outlet and require only a quick initial setup for continuous operation: they are simply plugged in and rotated to face the BEAD, providing wireless power up to 2m away.**^{15,16}

Page 10 Line 181

Using†The BEAD's onboard IMU's (Bosch, BMI270) embeds a proprietary decision tree model that continuously evaluates accelerometer signals to detect motion and perform walking/running activity ~~detection~~ recognition, generating hardware-level interrupts on activity change. **Using this interrupt allows the wearable device to enter a low-power mode that consumes 10x less power while maintaining quick wake timing (< 35 ms from low-power standby to normal operation, fig. S45) to collect real time gait data immediately once the wearer begins walking, with no additional effort or interaction required to begin data logging.**

Modifications to the Supplementary Information (Methods)

Page 24 Line 470

Step count was extracted from the angular velocity data of each thigh-located device (one gold-standard LEGSys device and one biosymbiotic device per thigh, four devices total). Step count extraction was automated using a MATLAB script and validated by a member of the study team; the MATLAB script isolated per-device all peaks in the angular velocity data with magnitude at least 50 degrees per second and inter-peak spacing at least 1 second apart.

Page 25 Line 477

Walking speed was determined from this test by manual analysis of the angular velocity data of each thigh-located device (one gold-standard LEGSys device and one biosymbiotic device per thigh, four devices total): the distance covered during the walk (3m from chair to cone + 3m from cone to chair) was divided by the time from toe-off of the first step after standing up to heel-strike

of the last step before turning around to sit down. These times were extracted by manual inspection of the angular velocity waveforms.

Page 27 Line 537

Training dataset creation

Angular velocity data from upper thigh placed devices were isolated from each original cohort subject's continuous 60-second walk dataset. Subject data were separated into individual steps and normalized using an automated MATLAB script, with all single steps aggregated into the training dataset ($n = 2057$ steps). The individual step snippets were manually reviewed to remove any erroneously tagged peaks or step boundary detection errors. The dataset was split 70-30 for training and validation. Labels were generated per-subject according to their FFP status determined from *in vivo* studies.

Page 28 Line 555

During training, 5-fold stratified cross-validation was employed and model parameters adjusted accordingly to mitigate model overfitting on training data and improve generalizability.

Page 28 Line 560

To mitigate the effect of class imbalance on the model, balanced class weights and randomly shuffled, stratified sampling of the training dataset were used during model training. Class weights were set to be inversely proportional to frequency of the class in the training set. This imposed a greater penalty on misclassification of the minority class (pre-frail) and encouraged better recall and f1-score.

Page 29 Line 574

On-device inference

When the BEAD's onboard IMU recognizes walking/running activity (through its proprietary embedded decision tree model analyzing its accelerometer signals), the IMU sends a hardware-level interrupt to the onboard microcontroller to wake it up. The microcontroller then enables the IMU's gyroscope and buffers ten seconds of 16-bit angular velocity data sampled at 200 Hz.

Individual step isolation is then performed by finding peaks in the angular velocity buffer (local maxima with at least 50 degrees per second angular velocity and ~ 1.1 seconds inter-peak spacing, parameters determined from the training dataset) and copying 218-sample step snippets centered at the index of each peak to a new buffer.

The sample times and angular velocity from a step are normalized $[0, 1]$ and transformed using the minimized, embedded MINIROCKET to extract features. The features are then passed to the on-device random forest classifier, with the output class timestamped and stored in on-device non-volatile memory. All steps detected in the buffer are analyzed sequentially. During this time, the IMU's gyroscope remains active and continues to collect samples. Once the microcontroller has processed all steps, it retrieves any samples from the gyroscope and continues feature extraction and inference until the IMU detects that the wearer has stopped moving. The microcontroller then performs any last inferences, disables the IMU's gyroscope to save power, and enters a low power sleep until motion is detected again.

Comment 6

Other comments

Include number of participants and other quantitative data in the abstract. What were the results? Did it work?

Our Response

We thank the reviewer for their comment and suggestion to improve legibility of the abstract. We have now added quantitative data and key results.

Modifications to the Manuscript (Abstract)

Page 1 Line 22

We utilize this framework to perform gait-based frailty assessment during *in vivo* trials ($N_1=20$) with results that match gold standard diagnostic tools. Clinical utility, ~~and~~ model stability, ~~and~~ on-device inference are validated through *in vivo* trials ($N_2=14$) and ten-day-long extended wear experiments, demonstrating continuous operation without wearer intervention and autonomous longitudinal analysis of high sampling rate biosignals.

Comment 7

Intro: Authors say "...requiring no interaction by the wearer at any point, including during recharging and data collection" However recharging and data collection are examples of user interaction. Where does the BLE go? How does that get back to the clinic? If the device is shipped how does the user setup the charger under their seat?

Our Response

We thank the reviewer for their comment and questions and now highlight the autonomous functionality of this platform.

To clarify, by user interaction we mean steps that a user typically needs to perform when using a standard brick-and-strap device. This includes taking the device off to shower, taking the device off to charge, either by plugging in a cord or using a near field charging dock, or using adhesives to attach the sensor (similar to a continuous glucose monitor) which need to be replaced every 3-5 days.

With the BEAD, the wearer does not need to take any actions outside their normal routine for the device to operate.

With our device, recharging is performed passively using far-field power harvesting: a commercially available wireless power caster placed at a work desk or bedside transmits power to the wearable up to 2 meters away when the wearer sits or sleeps in front of it. This power caster plugs in to any wall outlet and requires little setup; it simply needs to be plugged in facing the wearable device.

Data collection is performed on the device as the wearer takes steps, they do not need to push any buttons or launch a phone app. The data can be transmitted off the BEAD over BLE via an internet connected device (in most cases a smartphone) which can deposit the analyzed data (# of frail steps and normal steps) directly into the electronic health record.

It is important to note that with other devices data analysis typically has to be performed in the cloud. This requires sending all raw data with high resolution, stresses the connection point (i.e., drains the smartphone battery and consumes data), and requires computation in the cloud, which has its own associated resource limits and costs. Hence there is a substantial benefit in computing on the wearable itself.

We have amended the manuscript to more clearly make this distinction.

Modifications to the Manuscript

Page 6 Line 99

Therefore, wearable technologies capable of chronic operation^{22,23}, e.g., biosymbiotic electronics, are critical for establishing a new standard for frailty determination. Biosymbiotic ~~wearable device~~ **technology** utilizes soft, conformal 3D-printed mesh bodies, far-field power harvesting, and **autonomous** ultralow-power electronics to create ~~a~~ **sensing platforms** that ~~can~~ operate continuously without any interaction required ~~by~~ **from** the wearer.¹⁵⁻²⁰

Page 10 Line 173

Onboard wireless far-field power harvesting **highlighted** using commercially available, FCC-approved power casting systems enable wireless recharging at-distance, such as at a work desk or during sleep, without wearer interaction.^{15,16} **These power casting systems plug into any wall outlet and require only a quick initial setup for continuous operation: they are simply plugged in and rotated to face the BEAD, providing wireless power up to 2m away.**^{15,16}

Page 10 Line 181

~~Using~~ **The BEAD's** onboard IMU's (Bosch, BMI270) embeds a proprietary decision tree model that continuously evaluates accelerometer signals to detect motion and perform walking/running activity ~~detection~~ **recognition**, generating hardware-level interrupts on activity change. **Using this** interrupt allows the wearable device to enter a low-power mode that consumes 10x less power while maintaining quick wake timing (< 35 ms from low-power standby to normal operation, fig. S45) to collect real time gait data immediately once the wearer begins walking, **with no additional effort or interaction required to begin data logging.**

Page 11 Line 213

Analyzed data, i.e., timestamped inferences, can be offloaded via an internet connected device such as a smartphone, aggregated into total steps classified as frail versus healthy, and deposited directly into the electronic healthcare record (EHR), shown in fig. S7A.

Unlike with traditional brick-and-strap wearable devices, a patient is not required to continuously interact with the BEAD to perform data collection, see fig. S7B for the control flow that enables autonomous data collection. They also do not need to remove the device for charging or when bathing. Only a brief initial setup of the power caster is required for indefinite collection of high-fidelity biosignals, enhancing accessibility, user experience, and improving overall patient compliance.^{10,15}

Page 18 Line 346

Since the ~~biosymbiotic wearable device~~ **BEAD** requires no interaction from the patient **beyond initial placement of the powercaster at a desk or bedside**, is recharged wirelessly, and can be worn while bathing, the patient would simply wear the device on their thigh and continue with their normal routine. During habitual gait, i.e., **step activity recognized by the BEAD's onboard IMU**, the device wakes up, records **continuous angular velocity of the leg**, isolates individual steps, and performs inference, aggregating the results over time and transmitting them off-device when possible, potentially integrating them directly into **EHRs** / electronic patient health information (ePHI)²⁸ or transmitting them directly to the telehealth provider.

Page 22 Line 423

Biosymbiotic wearable technologies with on-device ML offer substantial advantages over existing wearable devices, allowing patients in low-resource settings to receive clinical-grade frailty assessment. The soft, breathable, lightweight form factor that defines biosymbiotic architecture is adhesive-free and tailored to conform to each patient for maximum comfort, allowing for weeks-long wear without irritation. Far-field power harvesting enables continuous operation without removal for charging (which is typically required by conventional brick-and-strap wearable devices), critical for potentially frail individuals with limited mobility. Exclusively on-device inference removes the requirement for constant connectivity that is often limited or unavailable, especially in rural communities, and eliminates the need for storing the large amounts of data on devices or costly cloud infrastructure, and allows for straightforward integration into EHRs.

Comment 8

Pg 3. These statements have no references.. “On-device continuous step isolation and inference significantly reduces the amount of data transmitted by the device, conserving battery life, enhancing data privacy, and providing quantifiable long-term metrics with an overall accuracy greater than 90% when classifying between healthy and pre-frail steps. Studies with healthy and pre-frail subjects aged 65 and older demonstrate fidelity on-par with current clinical gold standard gait analysis devices, and showcase device ability to perform real time, exclusively on-device inference for continuous frailty status determination”

Our Response

We thank the reviewer for their comment. These statements are our claims based on the results presented in the paper. Some of these statements have been altered in response to reviewer comments, however, the new statements are supported by the data presented in this work. We have modified the manuscript to more clearly indicate that these are our claims by including quantitative data and direct figure references.

Modifications to the Manuscript

Page 3 Line 74

On-device continuous step isolation and inference significantly reduces the amount of data transmitted by the device by nearly 99% (8 bytes for a timestamped step inference vs 436 bytes of raw data per step, Fig. 5D), conserving battery life with a 21% decrease in average power consumption (Fig. 3C), enhancing data privacy by minimizing data management requirements, and providing quantifiable long-term metrics with an overall accuracy greater than 90% when classifying between healthy and pre-frail steps (Fig. 4E). *In vivo* studies with healthy and pre-frail subjects aged 65 and older demonstrate gait measurement fidelity on-par with current clinical gold standard gait analysis devices (Fig. 2B and 2D), and showcase the BEAD's device ability to perform real-time (<330ms from raw data to inference result, fig. S8), exclusively on-device inference for continuous frailty status determination.

Comment 9

The authors say “The current clinical standard for frailty diagnosis, illustrated Fig. 1A, is reactionary rather than preventive, with diagnoses often only occurring after incidental fall or hospitalization.” The new proposed method as per Figure 1 is also reactionary, as the gait assessment appears to happen after a fall and initial consultation. All this new method would do is remove one follow up with an in-clinic assessment, with an in-home assessment. If anything this actually delays diagnosis as at home measurement takes longer than in clinic assessment, and will no doubt have some usability and setup issues with elderly folks.

Authors should add some downsides of wearables to provide a balance, e.g. skin attachment, slipping, rubbing/comfort, social stigma etc

Our Response

We thank the reviewer for their comment and suggestions to improve manuscript clarity and cohesion. While Fig. 1A was primarily intended to portray the current state of the art, it clearly is confusing for readers. The point of our device is that it can be used before adverse events because it simplifies diagnosis if there is a small probability of frailty in an accessible way. We have added and modified text in Fig. 1 and Fig. 5B to indicate the current standard of care and better highlight the proposed preventive approach, which would occur before a fall or hospitalization.

We want to point out that biosymbiotic devices are designed to mitigate the issues that current-generation wearables face, with comfortable, irritation-free, conformal form factors that stay in place and can easily be hidden under clothing, which we show in Fig. 5A.

Regarding usability and setup issues, biosymbiotic devices operate autonomously and by limiting all user interaction to one-time, simple placement of a power caster, we can reduce the barriers to diagnosis.

Modifications to the Manuscript

Page 9 Line 158

Current generation wearable devices face issues with subject interaction through charging requirements, technology management of smartphones or specialized devices, and secure placement and skin attachment, with patient discomfort and irritation often arising from rigid device bodies, abrasive textile or non-breathable plastic attachment straps, and adhesives which need to be replaced often.^{23,29} To avoid these common issues, BEADs employ biosymbiotic device architecture, which ensures lightweight comfort and the ability to operate continuously for weeks without input from the wearer.^{15,20}

Page 18 Line 339

The device introduced in this work, showcased Fig. 5A, provides a new approach for home-based frailty diagnostics in a low-profile, unobtrusive form factor which is easily hidden under clothing, shifting point of care away from medical clinics, and enhancing current telehealth offerings.

Modifications to the Manuscript (Figures)

Page 5 Line 85

Figure 1

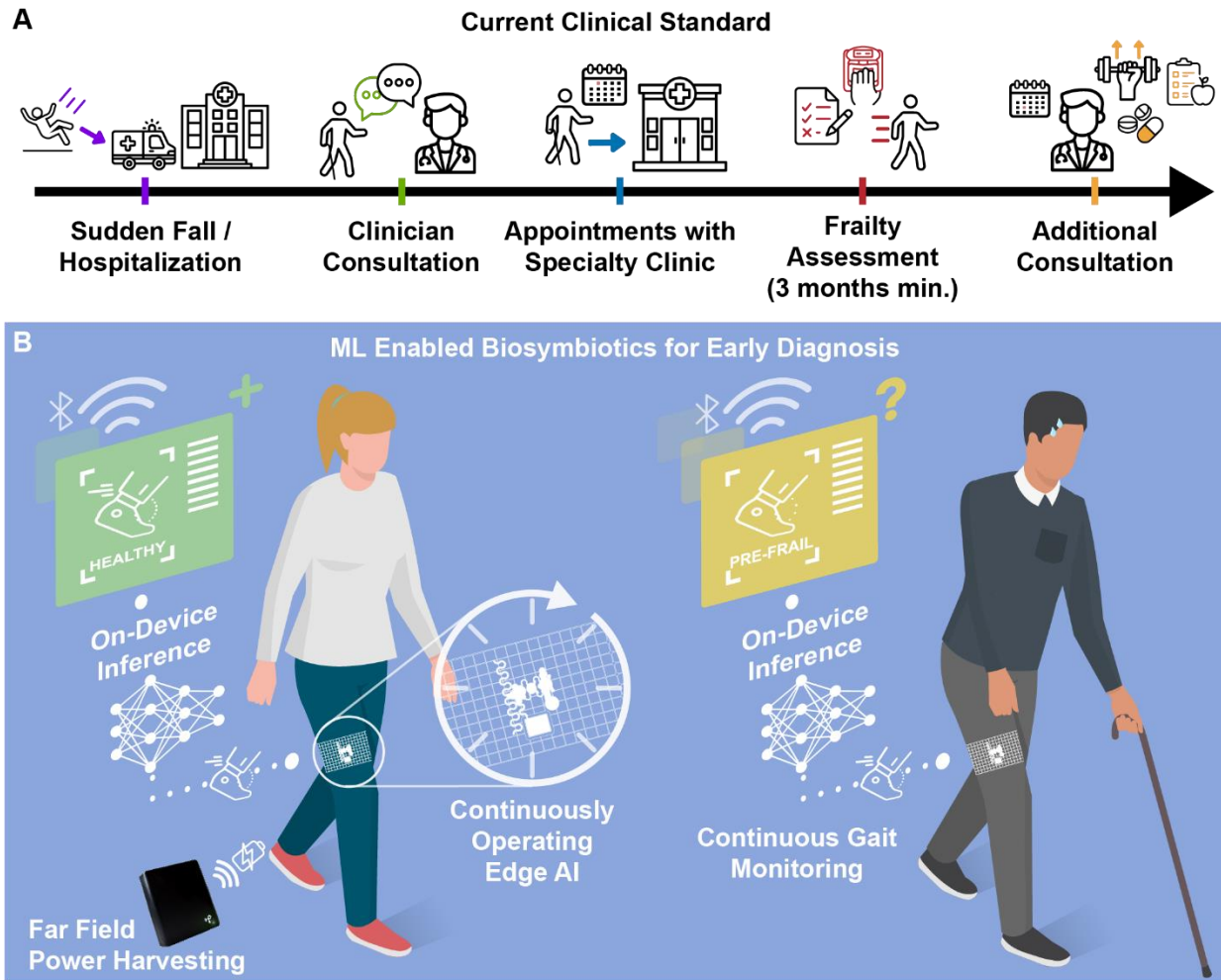


Figure 1. Continuously operating wearable AI for on-device frailty assessment. A. Illustration showing the reactive current clinical standard for diagnosing frailty. **B.** Implementation of wearable performing on-device inference from IMU recorded gait, wireless far-field charging of the wearable device, and Bluetooth Low Energy data output from wearable.

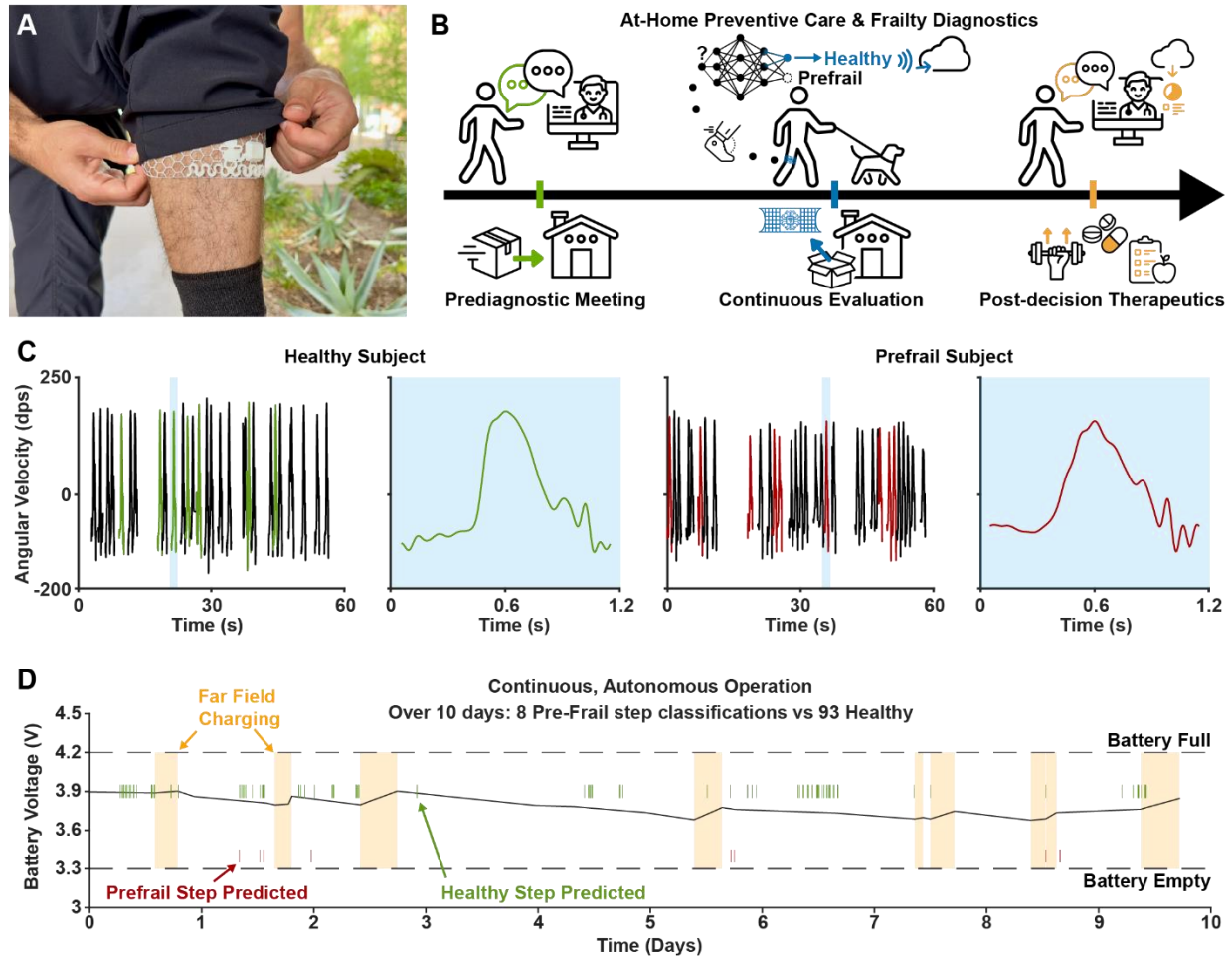


Figure 5. Continuous gait monitoring for frailty detection. **A.** Photograph demonstrating the low profile of BEADs that allows them to be worn discretely under clothing. **B.** New approach for home-based frailty diagnostics and preventive care. **C.** Plots showing results of step inference during 60s continuous walk from a healthy (left) subject and a pre-frail (right) subject, with insets highlighting an individual step from each subject. **D.** Battery voltage and inference results from ten-day continuous wear on healthy subject.

Comment 10

Results section

Gold standard.

When were results taken from Bio sensors? If the goal is to show they can last for long periods then the data should be collected after say 60 days and/or with low charge, not when fully charged.

Our Response

We thank the reviewer for their comment and suggestions to enhance the impact of this work.

We would like to emphasize that there have been multiple sets of experiments that were performed in this work.

The gold standard results were collected during *in vivo* studies ($N_1=20$) in a hospital setting. During the experiments for gait assessment comparison with gold standard devices, acceleration

and angular velocity data from leg- and trunk-mounted biosymbiotic devices were recorded continuously. The purpose of these experiments was simply to 1. demonstrate the sensing fidelity of the biosymbiotic platform, not continuous operation (which was the purpose of the 10-day-long extended wear experiment) and 2. collect data against the gold standard method of frailty detection to provide a training dataset for development of the on-device diagnostic prediction model. We have amended the manuscript to improve clarity on these points.

For chronic operation tests, the biosymbiotic devices are charged at distance using far-field power casters. If the wearer routinely returns to a charging point (like the bed or the work desk) the BEAD remains at a stable battery voltage (~3.7 to 3.9 V), which we show in Fig. 5D.

However, we agree that it is valuable to demonstrate reliable operation of the BEAD across varying levels of charge and have amended the supplementary information to include a new figure, fig. S12, which demonstrates sensing fidelity at different battery voltages, with results showing that the gait sensing fidelity remains unaffected by battery voltage.

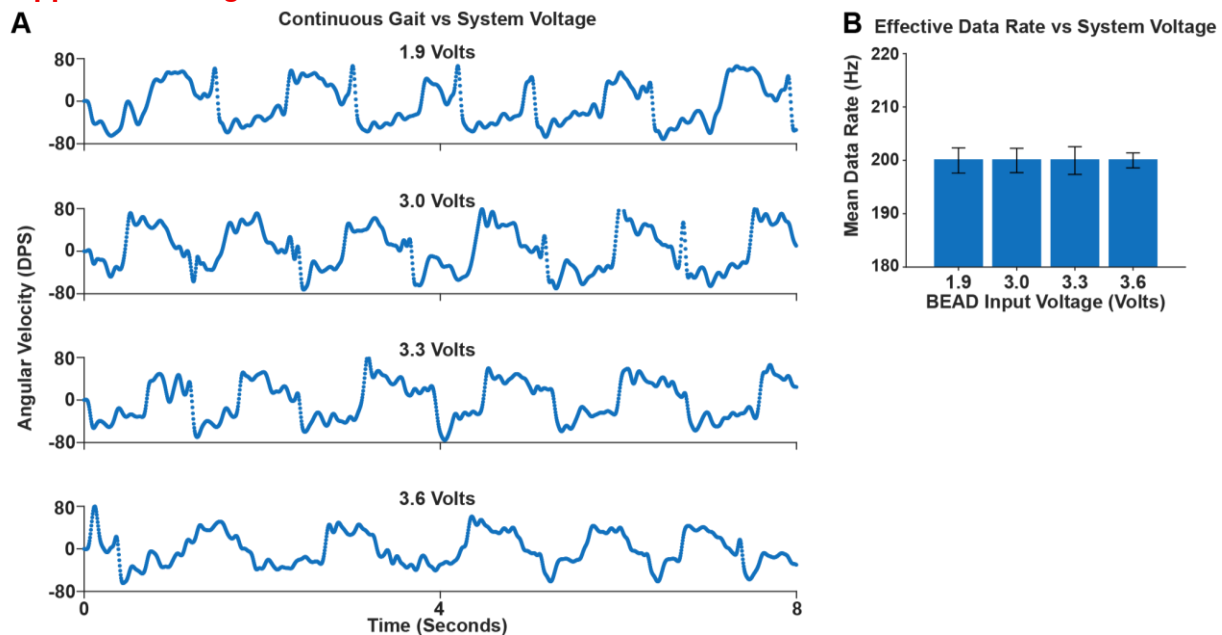
Modifications to the Manuscript

Page 19 Line 383

The experiment shows continuous ~~operation~~ **high fidelity digital gait assessment without impact by state of charge (fig. S12)**, reliable recharging, and long-term model stability throughout, with only ~8.6% false positive rate.

Modifications to the Supplementary Information (Figures)

Supplemental Figure 12



Supplemental Figure 12. Gait data fidelity is unaffected by battery voltage. A. Habitual gait angular velocity data logged over BLE at different system voltages. Samples plotted as discrete points. **B.** Effective recording data rate of the BEAD (from on-device IMU to microcontroller to laptop logging device over BLE) at varying system input voltages, plotted with standard deviations. Effective sampling rate consistently approaches 200Hz at all system input voltages. BLE transmission overhead, which shares CPU resources, introduces minor deviations in effective data rate.

Modifications to the Supplementary Information (Methods)

Page 30 Line 607

IMU data fidelity versus system voltage

The same source measurement unit, BEAD, and treadmill setup was used as during the walking speed versus current consumption test. A member of the study team wore the BEAD above the knee while walking on the treadmill at 1.8mph. Continuous gait data rate was measured while the output voltage of the SMU powering the wearable was adjusted to 1.8, 3.0, 3.3, and 3.6 Volts.

Comment 11

Methods

Seems like methods come after results section, doesn't make sense? Also some methods are in results section?

Our Response

We thank the reviewer for their comment and suggestion to improve manuscript clarity and legibility. The manuscript was submitted according to the Nature Communications style guide, its standard format is to have the Methods at the end, for reference and convenience we have included below a device-focused article^{R5} recently published in Nature Communications. In the main text we sometimes mention methods, such as fabrication or data analysis methods, when it is critical to the discussion of the results. We have revised the manuscript to move more methodology to the methods section and streamline discussions.

R5. Zhou, M., Yan, B., Yang, F. et al. A wireless optogenetic stimulation system for long-term function evaluation of mice forelimb with sub-nerve resolution. Nat Commun 16, 8702 (2025). <https://doi.org/10.1038/s41467-025-63746-y>

Modifications to the Manuscript

Page 13 Line 237

~~To create the dataset for training a machine learning model, a~~ Angular velocity from upper thigh located ~~placed~~ biosymbiotic devices, as sample data shown in Fig. 4A (left) and device placement shown in Fig. 2C, is normalized (Fig. 4A, middle) and used to create the training dataset Fig. 4A (right) ~~is isolated from each subject's continuous 60-second walk dataset.~~

Page 13 Line 244

~~Each subject's data are separated into individual steps and normalized, with all single steps aggregated into the training dataset (n = 2057 steps), shown Fig. 4A (right), and the dataset split 70-30 for training and validation. Steps are labeled according to each subject's FFP determined during in vivo studies.~~

Illustrated in Fig. 4B, ~~a time series classifier is trained off device using the open-source artificial neural network library Keras on a desktop workstation. A~~ a linear classifier model trained on a workstation is readily quantized and optimized into memory-efficient C code for deployment on resource-constrained embedded devices.

Page 20 Line 382

~~No step predictions are logged on day 3, as the wearer performed vigorous non-running sport during most of the day and led a more sedentary lifestyle in the evening.~~

Modifications to the Supplementary Information (Methods)

Page 27 Line 537

Training dataset creation

Angular velocity data from upper thigh placed devices were isolated from each original cohort subject's continuous 60-second walk dataset. Subject data were separated into individual steps and normalized using an automated MATLAB script, with all single steps aggregated into the training dataset (n = 2057 steps). The individual step snippets were manually reviewed to remove any erroneously tagged peaks or step boundary detection errors. The dataset was split 70-30 for training and validation. Labels were generated per-subject according to their FFP status determined from *in vivo* studies.

Page 28 Line 557

The trained model was quantized using the open-source artificial neural network library Keras and converted into ONNX (Open Neural Network Exchange) format, floating-point layers compressed/optimized using STMicroelectronics' X-Cube-AI toolset, and deployed onto a microcontroller.

Page 32 Line 638

The wearable device was not recharged during the weekend (days 3 and 4) or when the team member was out of office (day 6). No step predictions are logged on day 3, as the wearer performed vigorous non-running sport during most of the day and led a more sedentary lifestyle in the evening.

Comment 12

What is the functionality of the mesh fabrication, not clear what that is for and who it fits into the sensor design, manufacture and operation.

Our Response

We thank the reviewer for their comment and suggestion to provide more detail on the wearable biosymbiotic mesh. The 3D-printed, flexible thermopolymer mesh is designed using 3D scans, photogrammetry, direct measurements taken from the patient, or standard sizes in small, medium, and large. Each mesh can be 3D printed in under an hour and can be tailored to the individual, creating a perfect fit which is very lightweight, comfortable, and does not slip off or constrict the wearer's movements.

We have detailed biosymbiotic mesh fabrication in prior work we have now included this in our manuscript

Modifications to the Manuscript

Page 9 Line 158

Current generation wearable devices face issues with subject interaction through charging requirements, technology management of smartphones or specialized devices, and secure placement and skin attachment, with patient discomfort and irritation often arising from rigid device bodies, abrasive textile or non-breathable plastic attachment straps, and adhesives which need to be replaced often.^{23,29} To avoid these common issues, BEADs employ biosymbiotic device architecture, which ensures lightweight comfort and the ability to operate continuously for weeks without input from the wearer.^{15,20}

Page 9 Line 165

Biosymbiotic devices embed soft electronics in lightweight, breathable, water- and sweatproof, patient-tailored (using 3D scans, photogrammetry, or direct measurements) polymer meshes that are rapidly 3D-printed using FDA-approved skin-safe thermoplastic polyurethane (TPU) filament.¹⁵ The thin, low mass, form factor of the device (6mm height max, weight 15g with 30 mAh LiPo battery) and breathable mesh, ~~3D-printed using FDA-approved skin-safe thermoplastic polyurethane (TPU) filament,~~ allow for nearly imperceptible wear without slippage, adhesives, or irritation.^{15,19}

Comment 13

No clear description of electronics circuit and sensor design, i.e. controller specification/model, power storage, sensor architecture etc

Our Response

We thank the reviewer for their comment and suggestion to improve clarity around electrical design. We have amended the supplementary information to include a new figure, fig. S3, with a full electrical schematic. We want to note that Supplementary Figure S2 contains a labeled component diagram that indicates the controller specifications / model and sensing architecture.

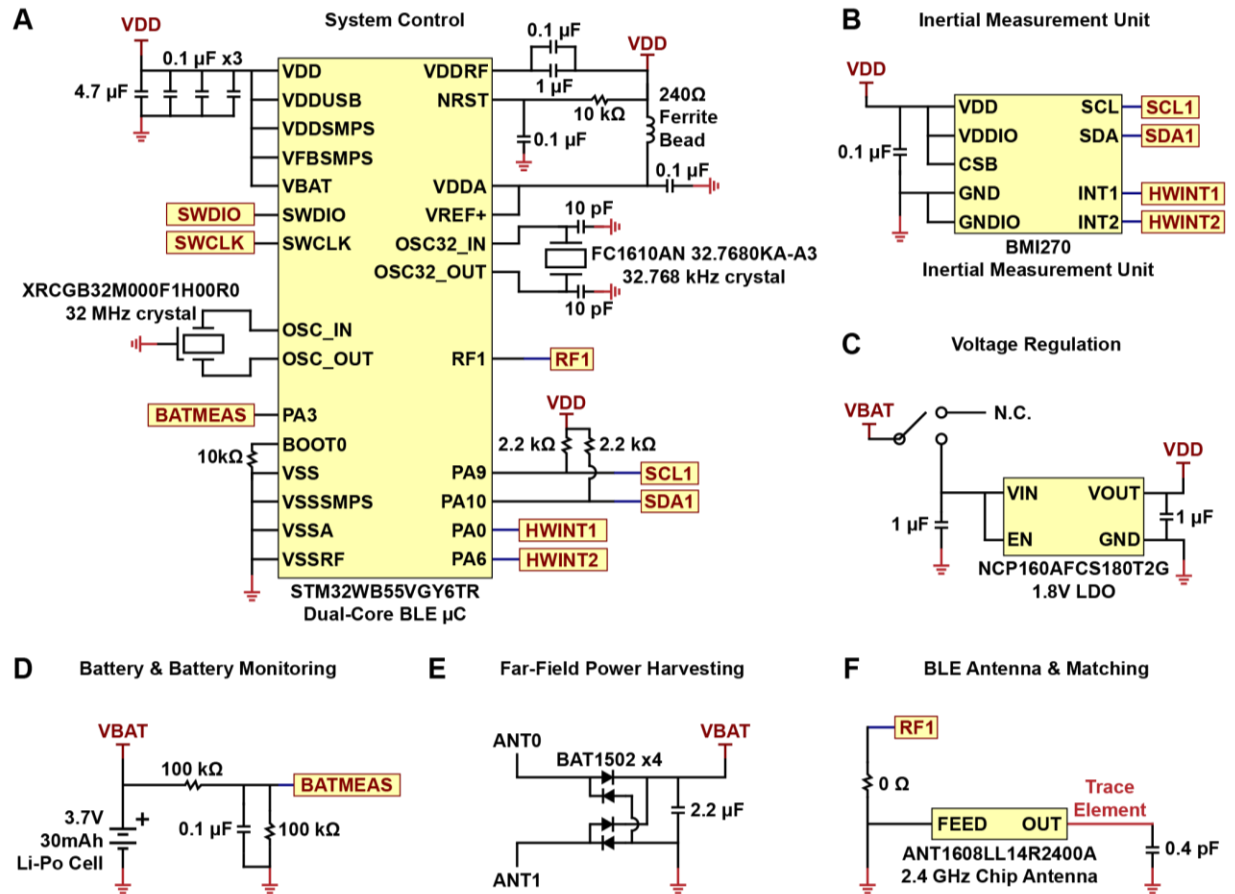
Modifications to the Manuscript

Page 10 Line 171

A functional block diagram, ~~and a detailed~~labeled component diagram, and an electrical~~system~~ schematic are included in Fig. 3B, ~~fig. S2,~~ and fig. S2~~3~~, respectively

Modifications to the Supplementary Information (Figures)

Supplemental Figure 3



Supplemental Figure 3. BEAD electrical schematic. Complete electrical schematic identifying subsystems with component values. **A.** System control with dual-core BLE microcontroller **B.** 6-axis IMU. **C.** Voltage regulation circuitry. **D.** Battery & battery monitoring circuitry. **E.** Far-field power harvesting / rectification with dipole antenna. **F.** BLE Antenna with 2.4 GHz chip antenna.

Comment 14

No clear description of algorithms for step count, walking speed and sit to stand detection. I found some description but it's in the results section

Our Response

We thank the reviewer for their comment and suggestions to improve clarity around step counting, walking speed, and sit-to-stand detection algorithms.

These metrics were collected during *in vivo* gold standard frailty assessment experiments. To train the on-device diagnostic prediction model, class labels, i.e., frailty states (healthy/pre-frail/frail), were required for subject data, which were generated using the Fried Frailty Phenotype and direct monitoring of *in vivo* subjects by members of the study team.

We have amended the text to explicitly detail the step counting and walking speed algorithms. Sit-to-stand detection was observed and timed directly by members of the study team.

Modifications to the Manuscript

Page 6 Line 110

Gait parameters from continuous 60-second walk test during *in vivo* trials ($N_1 = 20$ subjects, 16 healthy, 4 pre-frail) are shown in Fig. 2B. Step/stride time and step/stride variability are gait parameters typically assessed when performing frailty assessment in clinic, and can reliably distinguish frailty severity.^{3,5}

Page 24 Line 470

Step count was extracted from the angular velocity data of each thigh-located device (one gold-standard LEGSys device and one biosymbiotic device per thigh, four devices total). Step count extraction was automated using a MATLAB script and validated by a member of the study team. The MATLAB script isolated all peaks in the angular velocity data with magnitude at least 50 degrees per second and per-device inter-peak spacing at least 1 second apart.

Page 25 Line 477

Walking speed was determined from this test by manual analysis of the angular velocity data of each thigh-located device (one gold-standard LEGSys device and one biosymbiotic device per thigh, four devices total): the distance covered during the walk (3m from chair to cone + 3m from cone to chair) was divided by the time from toe-off of the first step after standing up to heel-strike of the last step before turning around to sit down. These times were extracted by manual inspection of the angular velocity waveforms.

Page 25 Line 482

Sit-to-Stand: Subjects began seated and were asked to stand up and sit back down without using their hands. This was repeated five times with their total time recorded by a member of the study team directly observing the subjects.

Comment 15

Seems like the sensor parameters (BME rate) were adjusted from the gold standard testing to 10-day tests, this doesn't make the gold standard representative of at home use.

Our Response

We thank the reviewer for their comment. We have detailed the purposes of the multiple experiments performed in this work in response to Comment 10 above. While the BLE packet rate was indeed changed between the gold standard comparison experiments and the ten-day continuous wear tests, this change has no relation to the system's sampling or inference rate, which occurs entirely on-device and does not rely on BLE, nor sensing fidelity. The 16-bit digital angular velocity data used during inference is collected entirely on the IMU at a fixed 200 Hz. This data rate is configured once during system boot and does not change at any point.

Unlike other existing wearables for gait-based frailty analysis that transmit raw data off-device over BLE for external or cloud processing, the BEAD performs gait data preprocessing and ML inference entirely on-device. During typical use the device only utilizes BLE to transmit small timestamped per-step inference results off-device, not large amounts of raw data. Modifying the BLE packet rate for this typical use case allows us to reduce energy consumption without affecting performance. We have amended the manuscript to clearly state this.

In the case that one needs to offload large amounts of raw data over BLE, such as during the gold standard comparison tests where in addition to the inference we also collect the raw IMU

data, the BLE packet rate can be increased to improve data offload rates and ensure responsive BLE behavior.

To improve clarity regarding data flow and BEAD system behavior, we have added an additional figure to the supplementary information, fig. S7 Data and system control flow, that details how angular velocity data is collected and processed to become a clinically relevant diagnostic indicator. We have modified the manuscript accordingly.

Modifications to the Manuscript

Page 11 Line 213

Analyzed data, i.e., timestamped inferences, can be offloaded via an internet connected device such as a smartphone, aggregated into total steps classified as frail versus healthy, and deposited directly into the electronic healthcare record (EHR), shown in fig. S7A.

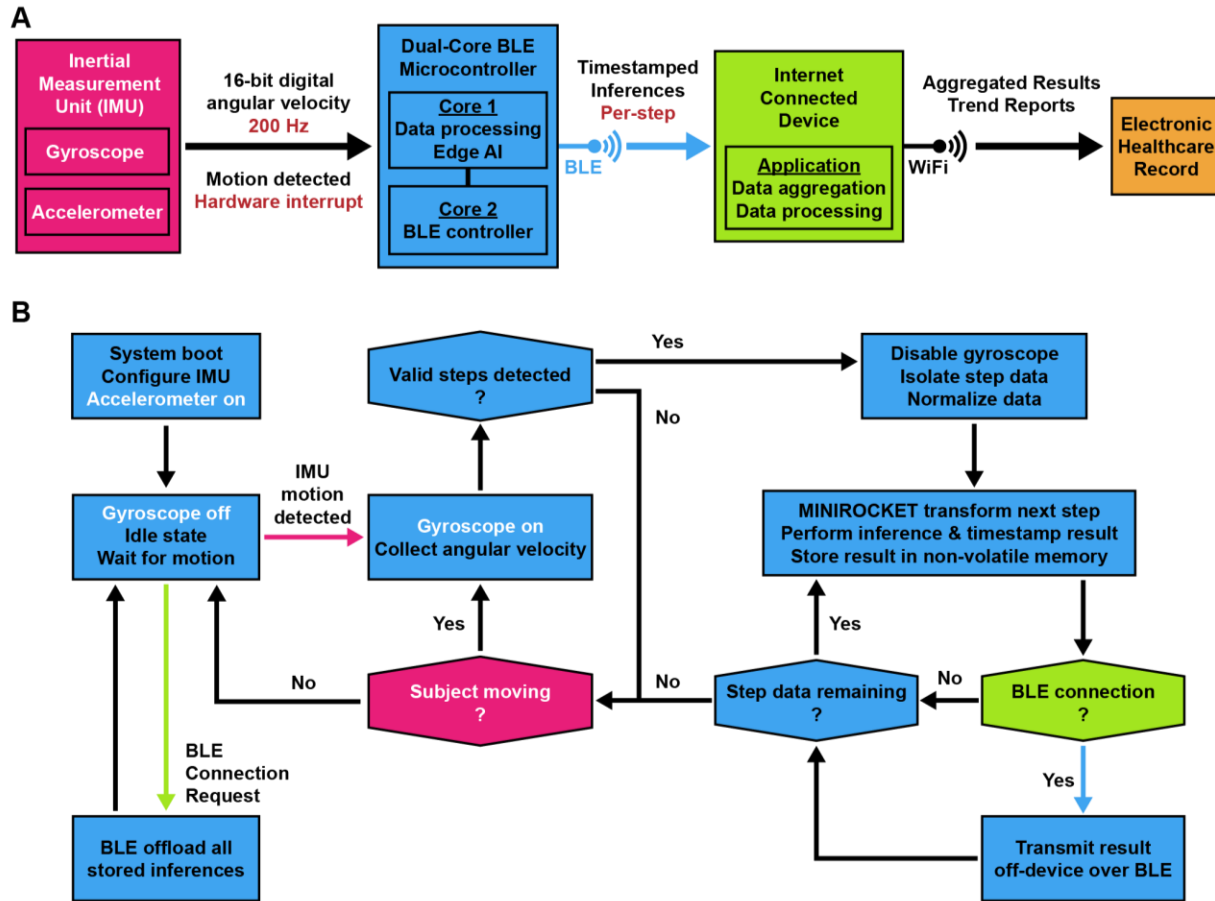
Unlike with traditional brick-and-strap wearable devices, a patient is not required to continuously interact with the BEAD to perform data collection, see fig. S7B for the control flow that enables autonomous data collection. They also do not need to remove the device for charging or when bathing. Only a brief initial setup of the power caster is required for indefinite collection of high-fidelity biosignals, enhancing accessibility, user experience, and improving overall patient compliance.^{10,15}

Page 19 Line 372

During the experiment, a BEAD worn by a healthy subject monitors their gait continuously, ~~waking when step activity is detected, collecting angular velocity data and performing inference on identified steps, and~~ storing only timestamped step inference resultsclassifications in nonvolatile memory on-device until data can be transferred off. In this typical use case, the BLE packet rate can be decreased to conserve power without affecting sensing fidelity, since all processing and inference occurs on-device without BLE. System performance is likewise unaffected by decreasing the packet rate, since only small packets of timestamped inference data are offloaded only when a logging device connects over BLE.

Modifications to the Supplementary Information (Figures)

Supplemental Figure 7



Supplemental Figure 7. Data and system control flow. **A.** Data flow showing how angular velocity data from the wearable device become clinically relevant diagnostic indicators. **B.** High-level control flow diagram detailing system behavior. Colors indicate the subsystem responsible for the control step and correspond to the identically colored subsystem from panel A. Hexagonal blocks indicate control decision points. White text indicates a specific activity occurs on the IMU.

Response to comments of Reviewer 2:

General Comments:

1. The manuscript highlights its main contribution as enabling a transition from reactive, post-hospitalization frailty diagnosis to preventive, home-based diagnostics. The topic should be of interest to the readers of NC, and the paper is generally well written. Some comments are provided below for consideration.

Our Response

We thank the reviewer for their comments and recognition of the manuscript quality. We have provided detailed responses to the comments below.

Comment 1

However, from Figure 1A it appears that patients are still prescribed the device only after a fall or hospitalization, following a clinical diagnosis. This raises a quick question: if such reactive clinical events still trigger device usage, how can the approach be truly considered preventive rather than reactive? The authors should consider clarifying under what circumstances the device would be deployed in practice, and whether there is a clear pathway for preventive adoption in at-risk but not yet hospitalized individuals.

Our Response

We thank the reviewer for their comment and suggestions to improve manuscript clarity and highlight novelty. The intent of Fig. 1A was to illustrate the current state of the art, however it was clearly confusing as depicted. To improve clarity, we have amended Fig. 1A and 1B, adding titles and text edits. We have also modified the text in Fig. 5B and the figure caption to indicate use of the BEAD for preventive care in addition to home-based monitoring.

We mention in our response to Reviewer 1 Comment 9 that the point of our device is for use before adverse events because it simplifies frailty diagnostics and makes attaining a diagnosis more accessible. This is achieved by deploying the device in an autonomous, continuously operating biosymbiotic form factor and limiting all user interaction to one-time simple placement of a power caster.

We have amended the manuscript to identify use cases where this system approach provides significant advantages over existing practices.

Modifications to the Manuscript

Page 23 Line 435

By reducing diagnostic barriers, these devices present compelling use cases in preventive medicine. For instance, continuous ECG monitors identify over 40% of arrhythmias only after day 3 of continuous recording,⁴⁹ allowing clinicians to intervene before sudden cardiac events⁵⁰ in cases that would otherwise be missed by short clinical assessments. This same principle could extend to ML-integrated biosymbiotic devices for frailty assessment in at-risk individuals, potentially reducing fall-related hospitalizations through early detection and intervention. BEADs offer a significant advantage over current approaches, which cannot provide scalable frailty diagnostics, let alone continuous monitoring.

Modifications to the Manuscript (Figures)

Page 5 Line 85

Figure 1

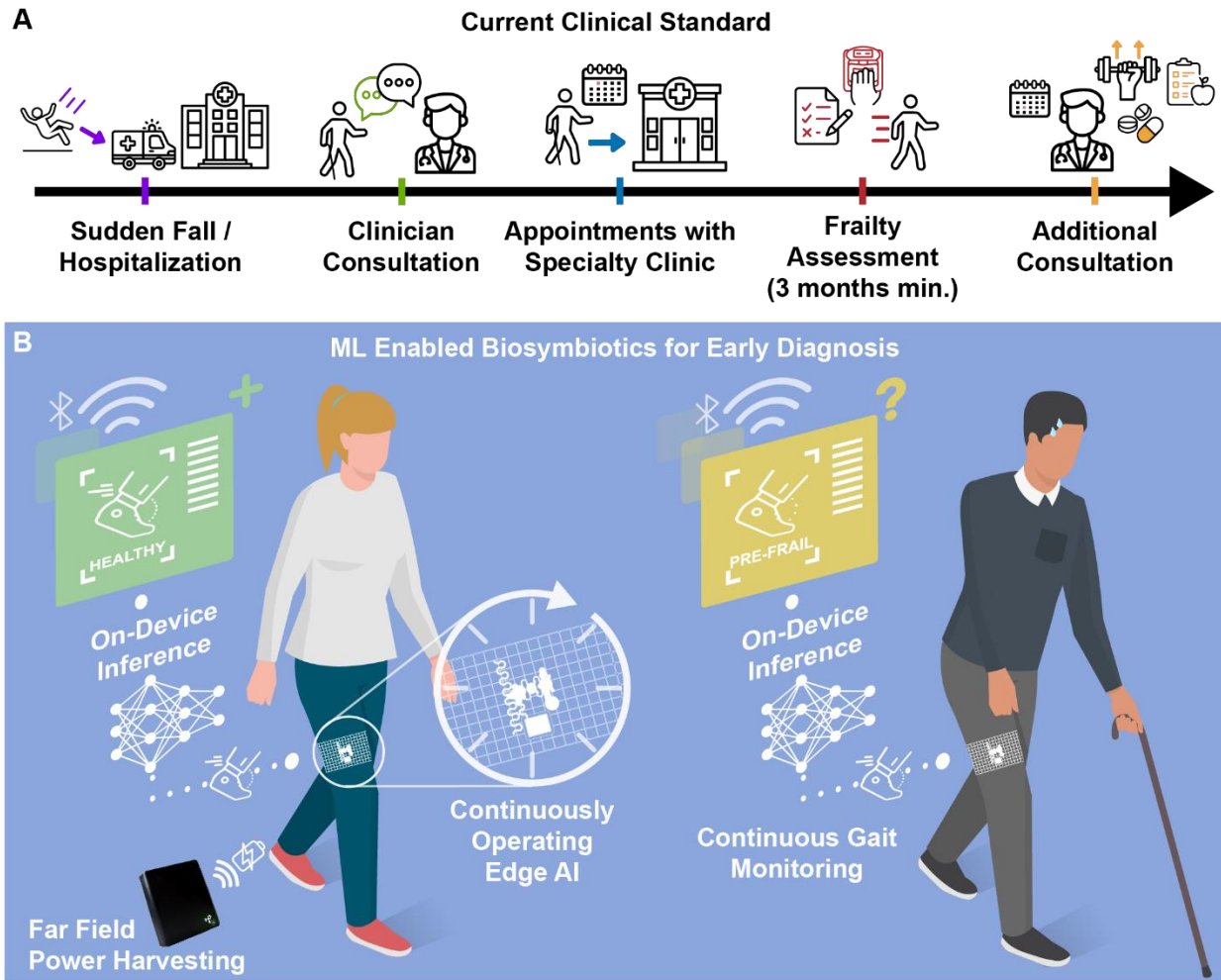


Figure 1. Continuously operating wearable AI for on-device frailty assessment. A. Illustration showing the reactive current clinical standard for diagnosing frailty. **B.** Implementation of wearable performing on-device inference from IMU recorded gait, wireless far-field charging of the wearable device, and Bluetooth Low Energy data output from wearable.

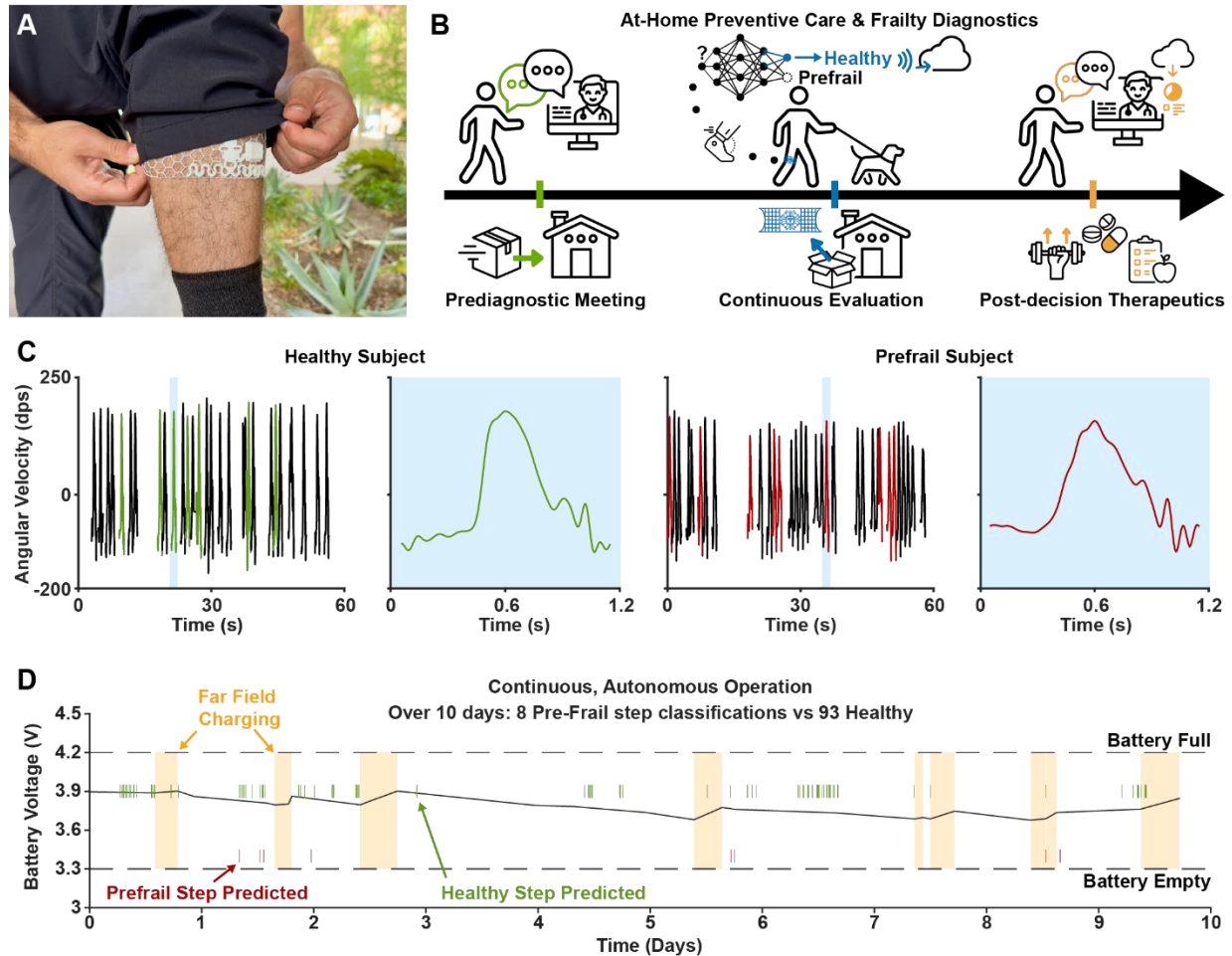


Figure 5. Continuous gait monitoring for frailty detection. **A.** Photograph demonstrating the low profile of BEADs that allows them to be worn discretely under clothing. **B.** New approach for home-based frailty diagnostics and preventive care. **C.** Plots showing results of step inference during 60s continuous walk from a healthy (left) subject and a pre-frail (right) subject, with insets highlighting an individual step from each subject. **D.** Battery voltage and inference results from ten-day continuous wear on healthy subject.

Comment 2

In the model training process, the dataset contains a strong imbalance between healthy and pre-frail subjects (16 vs. 4). Such imbalance may risk biasing the classifier towards the majority class. It is recommended that the authors explore strategies to mitigate this issue, such as stratified sampling, data resampling, or synthetic minority oversampling methods (e.g., SMOTE), to avoid skewing results and to better validate the robustness of the model.

Our Response

We thank the reviewer for their comment and suggestions to validate model robustness. While exploring various machine learning approaches, randomly shuffled, stratified sampling was utilized to help mitigate the impact of the class imbalance. Additionally, the final classification algorithm was implemented using a random forest classifier which took balanced class weights

into account during training, imposing a higher penalty on misclassification of the minority class. We have amended the manuscript to indicate this.

Subject scarcity is a limitation of this study, see our response to Reviewer 1 Comment 2. We want to highlight that the primary novelty of this work lies with the fully integrated BEAD platform itself, rather than the strength of the frailty assessment model, and that implementation of any edge AI with clinically proven diagnostic accuracy would require generating a substantially larger training set and therefore substantial resources to facilitate a multi-site and metropolitan area trial to collect relevant datasets.

We have amended the article to acknowledge subject scarcity and clarify our efforts in mitigating impact of class imbalance, including stratified sampling and random forest classification with balanced class weights.

Modifications to the Manuscript

Page 16 Line 310

While minor improvements may be gained through hyperparameter tuning or with a more advanced classification algorithm, clinical deployment will require substantially larger, more diverse training cohorts. Given local population dynamics (Tucson, AZ – 92,990 adults aged 65+) ³⁹, the general incidence of frailty in adults age 65 and older (~10-15%) ^{40,41}, and the ability of the local university-healthcare network to reach, recruit, and retain longitudinal study subjects, there become fewer than 200 potentially recruitable subjects in a large city. Considering recruitment rates of non-oncological clinical trials average as low as 0.85 subjects recruited per site per month, ^{42,43} statistically significant subject recruitment would require considerable resources, necessitating years-long, multi-institution, multi-site collaboration.

Modifications to the Supplementary Information (Methods)

Page 28 Line 557

The trained model was quantized using the open-source artificial neural network library Keras and converted into ONNX (Open Neural Network Exchange) format, floating-point layers compressed/optimized using STMicroelectronics' X-Cube-AI toolset, and deployed onto a microcontroller. To mitigate the effect of class imbalance on the model, balanced class weights and randomly shuffled, stratified sampling of the training dataset were used during model training. Class weights were set to be inversely proportional to frequency of the class in the training set. This imposed a greater penalty on misclassification of the minority class (pre-frail) and encouraged better recall and f1-score.

Comment 3

The work relies on a single 70/30 train-test split for model evaluation. While this provides an initial performance estimate, it does not fully rule out overfitting or ensure generalizability, especially given the small sample size. Cross-validation (e.g., leave-one-subject-out) or external validation would strengthen the claims regarding model robustness and clinical applicability.

Our Response

We thank the reviewer for their comment and suggestions to avoid overfitting and improve model generalizability. During training, stratified 5-fold cross-validation schemes were employed and model parameters were adjusted to mitigate overfitting to the training set. Adjustment of the MINIROCKET transform also took generalizability into consideration, with 420 convolutional

kernels (decreased from 10,000 kernels) providing consistent performance on the test set without degrading accuracy. We have amended the manuscript to include these methods. Regarding leave-one-subject-out cross-validation, please see our response to comment 5 below for more in-depth discussion.

Modifications to the Manuscript

Page 28 Line 553

A time series RF classifier was trained off-device on a desktop workstation using the transformed training dataset, and validated using the transformed test set. During training, 5-fold stratified cross-validation was employed to mitigate model overfitting on training data and improve generalizability.

Page 29 Line 568

Final parameters (420 kernels, 16 max dilations per kernel) were chosen to maintain average accuracy above 90% across shuffled test sets, ensuring high generalizability while mitigating overfitting to the training set.

Comment 4

Beyond cross-validation, the inclusion of learning curves would provide additional evidence regarding the model's bias-variance behavior and its capacity to generalize with more data.

Our Response

We thank the reviewer for their comment and suggestion to provide additional evidence toward model bias-variance behavior. We have amended the supplementary information to include a new figure containing the model's learning curves generated using two different cross-validation schemes, fig. S10. The results highlight the model's ability to learn without overfitting as the training set increases in size, however they also reveal bias in classification and the model fitting to subject-specific characteristics. See our response to Comment 5 below.

We have amended the manuscript to discuss these results.

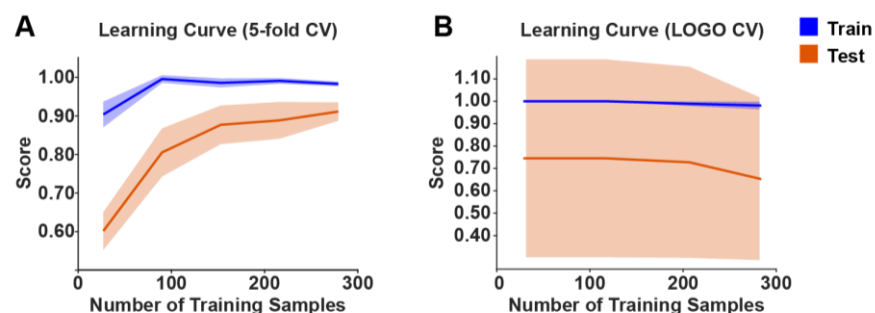
Modifications to the Manuscript

Page 15 Line 300

The f1-scores of the classification pipeline are therefore assessed and shown in Supplementary Table S2, with scores of 0.92 and 0.90 when predicting healthy and pre-frail, respectively, indicating excellent precision, recall, and ability to distinguish between classes. Figure S10 contains learning curves generated using two different cross-validation schemes: stratified, 5-fold (fig. S10A) and leave-one-group-out (fig. S10B), showing the model's ability to learn and generalize as the training set increases in size, although it is limited by the number of pre-frail subjects in the training set.

Modification to the Supplementary Information (Figures)

Supplemental Figure 10



Supplemental Figure 10. Model learning curves. **A.** Plot of learning curve of the frailty classification model using stratified 5-fold cross validation (CV), shaded region indicates standard deviation. Test accuracy increases with the number of training samples, showing model learning and improved generalizability, while train accuracy decreases, showing that the model is not overfitting. **B.** Learning curve of the model using Leave One Group Out (LOGO) cross-validation, shaded region indicating standard deviation. The results indicate that the model is learning subject-specific characteristics, since test accuracy decreases as the number of training samples increases.

Comment 5

Although the dataset includes a relatively large number of gait segments, these segments are not truly independent samples, as many originate from the same individuals. This could raise the risk that the model may be overfit to subject-specific characteristics, thereby limiting its generalizability to new individuals. Please discuss.

Our Response

We thank the reviewer for their comment and concerns about overfitting to subject-specific characteristics. Model training was performed using stratified cross-validation which did not take grouping (data points originating from the same group, or test subject) into account. We have generated the learning curve shown in fig. S10B (see our response to Comment 4 above) using leave-one-group-out (also known as leave-one-subject-out) cross validation. The model is fitting to subject-specific characteristics, since test accuracy is significantly worse and diverges from training accuracy (rather than converges) as training sample size increases. We mention in response to Comment 2 above that subject scarcity is a significant limitation of this study and have modified the manuscript to discuss these limitations. To develop a truly generalizable model likely 500 to 1000 independent subjects^{R6} are required, which would require at least 5+ sites of recruitment with excellent enrollment and adherence due to population statistics (see our response to Reviewer 1 Comment 2). This would require a large-scale effort beyond the scope of this proof of principle paper.

R6. Amjad, A., Qaiser, S., Błaszczyzyn, M. & Szczęśna, A. The evolution of frailty assessment using inertial measurement sensor-based gait parameter measurements: A detailed analysis. WIREs Data Min & Knowl 14, e1557 (2024).

Modifications to the Manuscript

Page 16 Line 309

They also reveal subject-specific overfitting that limits generalizability. While minor improvements may be gained through hyperparameter tuning or with a more advanced classification algorithm, clinical deployment will require substantially larger, more diverse training cohorts. Given local population dynamics (Tucson, AZ – 92,990 adults aged 65+)³⁹, the general incidence of frailty in adults age 65 and older (~10-15%)^{40,41}, and the ability of the local

university-healthcare network to reach, recruit, and retain longitudinal study subjects, there become fewer than 200 potentially recruitable subjects in a large city. Considering recruitment rates of non-oncological clinical trials average as low as 0.85 subjects recruited per site per month,^{42,43} statistically significant subject recruitment would require considerable resources, necessitating years-long, multi-institution, multi-site collaboration.

Comment 6

While the manuscript provides some waveform-level interpretations, the discussion remains largely algorithm-centric. A clearer articulation of how per-step inferences can be translated into clinically meaningful diagnostic indicators or longitudinal trend reports would strengthen the clinical relevance of the work. Please think about and discuss this.

Our Response

We thank the reviewer for their comment and suggestions to enhance clinical relevance, which we believe improve the impact of the article. We have amended the manuscript to emphasize the merit of per-step inference across longitudinal datasets and its utility as a clinically relevant diagnostic indicator.

Modifications to the Manuscript

Page 3 Line 58

Since conventional statistical analysis approaches for gait-based frailty assessment, such as examining stance time, step length, or frequency component analysis,¹¹ are difficult to automate and tailor to individuals,^{12,13} these large datasets provide a strong use case for machine learning (ML)-based data analysis, which can rapidly and accurately identify features that may be difficult for humans to recognize in a timely manner,¹⁴ especially repeated over longer time scales. ML approaches also capture complex, non-linear interactions which can be missed with conventional approaches, scale well and offer flexibility with the potential to learn from data.¹³ Performing on-device ML-based data analysis significantly condenses these large datasets, transforming weeks of continuous analysis into easily digested trends that can be transmitted with minimal bandwidth and data management requirements, shifting point-of-care from hospitals and clinics to homes through telehealth, reducing financial burdens on patients and decreasing strain on healthcare systems.

Page 20 Line 386

Combined, these results demonstrate that the ~~capacity of edge-AI on a continuously operating device to transform frailty diagnostics using gait-based assessment at home~~-power of per-step inference is realized through temporal ML analysis, i.e., statistical analysis of per-step inferences recorded from routine life over weeks to months. Cross-sectional analysis, i.e., statistical analysis of per-step inference results from short 60-second walks in labs or clinics once a month, is less representative and less accurate than temporal analysis.⁴⁶ Extracted through temporal ML analysis, the diagnostic indicator is the percentage of steps classified as frail (or pre-frail) (PSCF) relative to steps classified as healthy over the study duration. Ratios can be easily integrated into EHRs and read by a clinician at a glance, providing insight on a patient's status.

In cases where a patient's PSCF is not decisive, deeper analysis can generate longitudinal trend reports to enhance understanding. Since each inference result is timestamped, insightful metrics such as the number of frail steps per day providing a complete profile of when diagnosis occurs. Combining clinical history, patient lifestyle, and ML-based symptom analysis drives precision medicine and personalized care, likely improving patient outcomes and reducing healthcare cost.⁴⁷

Comment 7

The evaluation of the reported ~21% energy savings for on-device inference is confined to short-term, laboratory-based measurements. In real-world conditions across varying walking speeds, activity types, and usage durations, the device's overall power consumption may differ substantially. A more comprehensive energy analysis under daily living conditions is needed.

Our Response

We thank the reviewer for their comment and suggestion to provide more real-world results. We have amended the supplementary information to include a new figure, fig. S6, showing energy consumption using off/on-device inference across several different walking speeds. These results show energy savings have a direct correlation with walking speed, i.e., as the number of steps performed over a period increases, so does energy consumption, since the IMU activates the gyroscope more often. At or above 3.86 kph (2.4 mph), the gyroscope is practically enabled continuously and current consumption plateaus. Activity-based gyroscope activation can be manually controlled to limit the number of inferences per day and further save power.

Regarding activity types, the on-device IMU only enables the gyroscope when it recognizes walking or running activity through its proprietary embedded decision tree, thus any other activities are ignored automatically without additional power waste. When the gyroscope is active for angular velocity data collection, its baseline current draw (600+ μ W) dominates over the microcontroller, which idles under 250 μ W. We have amended the manuscript to discuss these changes.

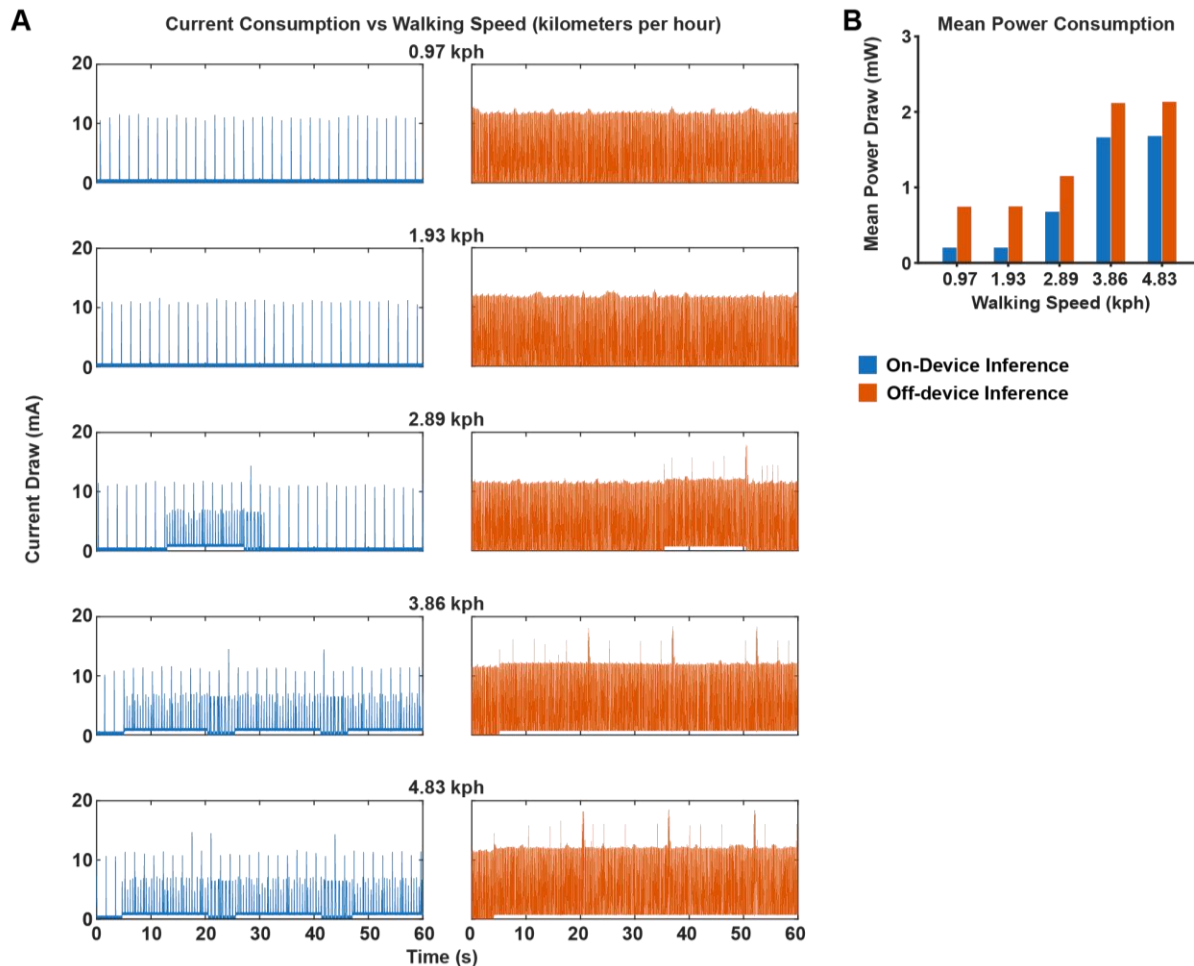
Modifications to the Manuscript

Page 10 Line 189

Figure 3C illustrates current consumption for a BEAD performing on-device inference versus transmitting raw step data for off-device analysis. Integrating over the entire 14 s duration reveals a 21% decrease in energy consumption (19.68 milliJoule versus 24.97 milliJoule, respectively) for single-step inference. Shown fig. S6, system energy consumption depends on walking speed, with the IMU's motion detection interrupt triggering more reliably at higher step rates. In response to the interrupt, the microcontroller activates the IMU's gyroscope, which has high power demands relative to the low-power idle state of the BEAD. Walking speeds below 2.89 kph (1.8 mph) did not reliably trigger the IMU's activity detection, thus no changes in system current consumption are noted for 0.97 and 1.93 kph (0.6 mph and 1.2 mph, respectively). At or above 3.86 kph (2.4 mph), the gyroscope is practically enabled continuously and current consumption plateaus. By employing more conservative inference approaches, e.g., limiting the number of steps classified for a given duration, one can maintain low energy consumption. Since the BEAD is worn for extended durations and ~~This enhances the utility of far field power harvesting (extended analysis time for the same far field charging duration) and reduces reliance on communication infrastructure as the resource-constrained wearable~~ can store inference results (<108 bytes/step) ~~over extended durations~~ when external logging or processing devices are not available, rate limiting inference is not as prohibitive as during data collection from short, dedicated, in-clinic sessions. ~~Additionally, this reduces the number of BLE connection intervals per step inference by at least two events. BLE can practically transmit up to 247 bytes/packet and directly outputting the inference result requires <10 bytes with timestamp and classification confidence, while raw angular velocity data for a single step approaches 500 bytes at 16-bit resolution and 218 samples/step, saving 0.079 mJ/inference generated. Practically this means that~~ Results from nearly 27,000 events can be stored on device, enabling 7-14 days (depending on daily step count) without

the need to activate BLE, providing exceptional flexibility in firmware design and increased reliability without the need for large onboard memory.

Modification to the Supplementary Information (Figures) Supplemental Figure 6



Supplemental Figure 6. Walking speed versus current consumption. **A.** Continuous BEAD current consumption performing inference on-device (blue) and off-device (orange) at 0.97, 1.93, 2.89, 3.86, and 4.83 kph (0.6, 1.2, 1.8, 2.4, and 3.0 mph, respectively). Off-device inference assumes consistent BLE connection. All plot X-axes are from 0 to 60s and Y-Axes from 0 to 20 mA. **B.** Mean current draw for each of the walking speeds tested, showing current consumption increases as walking speed increases. Walking speeds below 2.89 kph did not reliably trigger the onboard IMU's activity detection. Walking at 3.86 kph or faster led to continuous activation of IMU activity detection, with higher speeds not increasing current draw.

Modification to the Supplementary Information (Methods) Page 30 Line 592

Walking speed versus current consumption

Current consumption by a BEAD was monitored using a source measurement unit (SMU) (STMicroelectronics, STLINK-V3PWR). The output voltage of the SMU was adjusted to 1.9V, the 1.8V operating voltage of the BEAD with overhead for LDO dropout voltage. The SMU recorded sourced current with 50 kHz sampling rate. A member of the study team wore a BEAD

above the knee and stood adjacent to a treadmill (ASUNA, 8730). The treadmill speed was set to 0.6 mph and the study team member stepped on to the treadmill and walked at speed for 60 s. The team member then stepped off the treadmill for ten seconds while the speed was adjusted to 1.2 mph. The team member then stepped on to the treadmill and walked at speed for 60 s. This procedure was repeated for 1.8, 2.4, and 3.0 mph.

On-device inference was triggered automatically by the BEAD IMU's activity detection, with results logged to on-device non-volatile memory. For off-device inference, the same setup was used with a BLE connection established before the initial 0.6 mph test. During off-device inference measurements, the BEAD transmitted continuous raw angular velocity data off-device to a laptop.

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